

Random Undersampling on Imbalance Time Series Data for Anomaly Detection

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ABSTRACT

Random Undersampling (RUS) is one of resampling approaches to tackle issues with imbalance data by removing instances randomly from the majority class. Anomaly is considered as a rare case, thus the number of instances in the anomaly class is usually much lower than instances in other classes. In anomaly detection of time series data, an anomaly is identified when an unusual pattern exists. Duplicating the unusual pattern may lead to overfitting, which is why this study considered an undersampling method over oversampling approach. This study applied RUS on data with several algorithms to observe its effectiveness on different types of classifier. To prove the overfitting and underfitting issues, different ratios of training and testing were used. Five different evaluation metrics were considered to evaluate the performance of the approach used. It was found that RUS could improve the classification performance of every classifier and the best result was shown when RUS was applied on a deep learning algorithm.

CCS CONCEPTS

• **Computing Methodologies**; • **Machine Learning**; • **Learning Paradigms**; • **Supervised Learning**; • **Supervised Learning by Classification**;

KEYWORDS

Machine learning, Electricity theft detection, Anomaly detection, Imbalance time series data, Oversampling, Undersampling

ACM Reference Format:

Mulyana Saripuddin, Ms, Azizah Suliman, Assoc Prof DR., Sera Syarmila Sameon, Mrs, and Bo Nørregaard Jørgensen, PROFESSOR, PhD. 2021. Random Undersampling on Imbalance Time Series Data for Anomaly Detection. In *2021 The 4th International Conference on Machine Learning and Machine Intelligence (MLMI'21)*, September 17–19, 2021, Hangzhou, China. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3490725.3490748>

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MLMI'21, September 17–19, 2021, Hangzhou, China

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ACM ISBN 978-1-4503-8424-7/21/09...\$15.00

<https://doi.org/10.1145/3490725.3490748>

1 INTRODUCTION

The occurrence of imbalance data is common in most real-time datasets [1]. A time series dataset is said to be imbalanced when a class is dominant over another class. Imbalance can be evaluated using entropy method or ratio between numbers of instances in classes. In binary problems such as anomaly detection, using the ratio between the number of instances in the majority class and the number of instances in the minority class is the practical way to define the level of imbalance. Imbalance data is considered one of the top challenges in data mining [2]. A classifier tends to make biased predictions on imbalance data due to the lack of information presented by the minority class [3]. In the case of anomaly detection where anomaly cases are often rare, the classifier will tend to classify an anomaly as a normal case instead. Handling imbalance data in anomaly detection is highly crucial since domains such as fraud detection [1] and medical diagnostics [4] usually have a very low number of anomaly cases, which are often represented as the minority class. [5] mentioned that there are three top approaches to handle imbalance dataset. They are resampling techniques at data-level, feature extraction at pre-processing, as well as cost-sensitive learning at algorithm level. Of these three approaches, resampling has been widely applied to work with imbalance data.

The main purpose of resampling technique when managing data is to re-balance and reduce the skewness of data. This can be performed by either oversampling, undersampling or the combination of both. Oversampling involves increasing the number of instances in the minority class through duplicating existing data or creating new synthetic data from the existing instances. Undersampling involves the action of removing existing instances from the majority class until it reaches a specified ratio with the minority class. A combination of both methods is also sometimes considered [6]. However, oversampling leads to a clear issue as it may cause overfitting, especially random oversampling whereby it duplicates existing instances. Other oversampling approaches such as SMOTE and ADASYN create new information based on existing instances where the challenge is to verify whether the new information is valid or not. This study believes that in the case of time series data, normal instances usually have similar patterns to each other, which makes it the main reason to apply undersampling over oversampling. By applying random undersampling (RUS), it can reduce overfitting caused by iterative training on similar patterns.

Detecting anomaly/fraud users in electricity consumption is challenged with the existence of imbalance data in which the frequency of fraud users is often 'rare' as compared to normal users. Simply relying on a classifier's overall accuracy metric is also unreliable in the case of anomaly detection since the classifier will tend to classify cases as normal (bias). Furthermore, the cost of misclassification, i.e., False Negative (FN) and False Positive (FP), will put both consumer and utility provider at a disadvantage. In anomaly detection, overall accuracy is irrelevant to measure the performance of the classifier. Using several appropriate metrics at the same testing phase is required to better evaluate the classifier's performance in anomaly detection. Metrics can be used such as Area Under Curve (AUC), Recall, Precision and F-Measure [1].

The remainder of this paper is organized as follows: Section II discusses on the related works that have been reviewed to identify the research gaps and to validate on the needs of this study. Section III explains the Machine Learning (ML) algorithms and their relevance to this study while in section IV, the research design including description of dataset will be elaborated. Section V includes the experiment results together with the evaluation metrics used to evaluate the performance of techniques applied.

2 LITERATURE REVIEW

This section will elaborate on the studies conducted previously that are related to handling imbalance data. Several techniques of handling imbalance data are identified from this literature review. The selected literature review does not strictly focus on electricity theft detection only but generally focuses on the relevant techniques for handling imbalance data for any purpose. Electricity consumption is mostly presented in time series data whereby the number of fraud cases is usually much lower than normal cases. Most of the studies found such as [7] and [8] applied SMOTE, which is an oversampling method to handle imbalance data issues. [9] claimed that the synthetic data generated by the SMOTE method might lead to overfitting. They proposed ADASYN, which is an advanced version of SMOTE, as another approach to handle imbalance data issues on their time series data for anomaly detection. Their study confirmed that ADASYN performed better than SMOTE but no undersampling was included in the evaluation.

[3] compared the performance between SMOTE and RUS on ensemble classifier. The experiments showed that RUS performed better in increasing AUC and specificity. When applied with proper cross validation, SMOTE's performance could be increased. However, the dataset used in this study was not a form of time series dataset; therefore, it has to be confirmed whether RUS can really perform well on imbalance dataset including time series data.

A study conducted by [10] evaluated the impact of RUS on Random Forest classifier on big data. Four big datasets were used after modification to reach several ratios of imbalance. The datasets used were for the binary classification problem. However, none of them were time series data. The results of the experiments showed that increasing percentages of the minority class, which was similar to oversampling, could boost classifiers' performance. Nevertheless, partially undersampling the data also improved the performance. The performance depended on the characteristic of dataset used.

[11] adapted RusBoost, which is the combination of RUS and Adaboost, to handle imbalance data to detect non-technical loss (NTL) in electric distribution. The study claimed that unlike oversampling, RUS could reduce the load of computation by reducing training size and the likeliness of overfitting since repetition of similar examples was reduced. One month of average daily consumption was used as a feature of the dataset, which could be considered as a form of time series data.

Other than classification problems, resampling technique is also applied for imbalance time-series forecasting as conducted by [12]. This study compared random oversampling, SMOTE, and RUS on several baseline methods. The results confirmed their hypothesis that resampling strategies could significantly improve accuracy for imbalance time series forecasting.

[13] proposed RUS to tackle imbalance data issues and it seemed to improve the AUC score significantly. There are many other studies performed for anomaly detection including [14], [8], and [15]. However, they did not explicitly explain if the imbalance data issue was addressed or not.

From the related works, it is found that both oversampling and undersampling are relevant resampling techniques to be applied on certain types of dataset. SMOTE has been widely proposed by many works, but to the best of the authors' knowledge, none of them has evaluated the impact of RUS on different types of classifier for anomaly detection in imbalance time series data.

3 MACHINE LEARNING ALGORITHM

This section will briefly discuss on several ML algorithms that are used in this study, namely Decision Tree (DT), AdaBoost, Artificial Neural Network (ANN), and Deep Artificial Neural Network (DANN). DT represents the rule-based ML classifier, AdaBoost for ensemble classifier, ANN for neural network, and DANN represents deep learning. The relevance of considering these algorithms over other ML algorithms in this study will be explained later.

3.1 Decision Tree

In Decision Tree (DT), the algorithm makes decisions based on split branches to classify the lead node. The data are sorted down from the root of tree into the leaf node. The process is repeated into sub-root or simply understood as a top-down process. DT is useful when prediction is made on high dimensional data. In the case of time series classification, the dimensionality can be affected when data is widened to a specified period of time interval. [1] claimed that Decision Tree performed the best at handling big imbalance data. However, the study did not focus or involve any time series data. Therefore, the DT algorithm is considered in this study to validate whether it can perform well on highly imbalance time series data as well. In addition, [16] mentioned that there were several studies that previously used DT to perform NTL detection and this algorithm is regarded as one of the highly promising supervised learning algorithms.

3.2 AdaBoost

AdaBoost is a short form of Adaptive Boosting, a form of ensemble boosting algorithm. [1] theoretically explained that ensemble

learning could improve performance for classification on big imbalance data. AdaBoost in this study combined 100 DT as weak classifiers into a single strong classifier. DT was previously tested on imbalance time series data without resampling, whereby it failed to classify any sample into the anomaly class. As a result, DT was used as the weak classifier in AdaBoost. The summary of equation for AdaBoost used in this study can be represented as:

$$F(x) = \text{sign} \left(\sum_{n=1}^N w_n f_n(x) \right)$$

Where $N = 100$ as the number of weak classifier, f_n as the n_{th} classifier, and w_n is the corresponding weight. Given the dataset contains r points, $x_i \in \mathbb{R}^d$, $y_i \in \{0, 1\}$, where 0 denotes the negative or normal class and 1 denotes the positive or anomaly class. The weight of each data point is initialized as:

$$w(x_i, y_i) = \frac{1}{n}, i = 1, \dots, n.$$

3.3 Artificial Neural Network

ANN is an algorithm that computes input to output by adopting a biological neural network in animal brains. Nodes known as neurons are connected to transmit signals to other neurons in other layers. [16] mentioned that ANN has been widely used to perform NTL detection for binary case. Therefore, ANN is adapted in this study. To be specific, at hidden layers, Rectified Linear Unit activation function (ReLU) is used for faster learning and performance. Input data is defined as $X : x_1, x_2, \dots, x_m$, where m is 1,034 of days in the dataset with $W^1 : w_1^1, w_2^1, \dots, w_m^1$ as the weight set. The result can be attained as below:

$$y = \sum_{i=1}^m w_i x_i + \text{bias}$$

y is then fed into an activation function. The activation function used at the output layer is the Sigmoid function defined as:

$$f(y) = \frac{1}{1 + e^{-y}}$$

The learner is optimized by using Adam's optimizer and the loss is represented by using Binary Cross Entropy to set up the binary classification problem, which is defined as:

$$CE = -t_1 \log(f(y_1)) - (1 - t_1) \log(1 - f(y_1))$$

3.4 Deep Artificial Neural Network

In this study, DANN is improved from ANN by increasing the number of hidden layers. Deep learning is believed to be able to capture time shift and sequential pattern in time series data as compared to other base classifiers. Four hidden layers were added into the existing ANN. All other parameters, activation function, optimizer, and lost function were maintained as in the ANN model described previously.

4 RESEARCH DESIGN

This section elaborates on the research design before feeding data into the ML algorithm that has been explained previously. The dataset used will be described before the explanation on the data pre-processing.

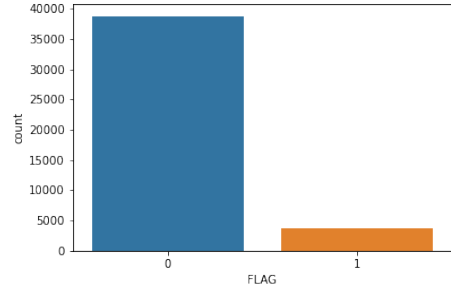


Figure 1: Number of Instances in Each Class For SGCC Dataset

4.1 Dataset

The dataset used in this study was the electricity consumption dataset provided by the State Grid Corporation of China (SGCC). This dataset has been widely used by many studies to detect electricity theft [7, 9, 14, 17]. The description of this dataset is provided in Table I. Based on the preliminary analysis conducted by [14], there was periodicity among normal consumers, which showed that there was a similar pattern among them

This dataset contained the electricity consumption data of 42,372 electricity customers within 1,035 days, from 1st January 2014 to 31st October 2016 with theft only representing nearly 9% of the dataset [14]. Figure I illustrates the different numbers of instances in both classes. Flag=0 is for normal consumers and Flag=1 is for theft.

4.2 Data Pre-Processing

To handle missing values in the dataset, the not a number (NaN) value was imputed by using the mean of row since every consumer was represented by each row in the dataset. Common in electricity consumption datasets, it combined the large and small power consumers altogether. Therefore, it is important to scale the data to bring them into common ground. MinMax scaling technique was applied, which can be defined as:

$$f(x_i) = \frac{x_i - \min(x)}{\max(x) - \min(x)}$$

Scaling the data is important since some algorithms are sensitive to diverse data, especially neural network algorithms. Undersampling the data is challenged with underfitting issues especially if important information is removed from the data. Therefore, different ratios of training to testing were used in this study. The ratios used were 90/10, 80/20, 70/30, and 60/40.

5 RESULTS

This section will present the overall results of experiments to observe the impact of RUS on imbalance time series data with the selected ML algorithm. Before that, the evaluation metrics that were considered will be explained since it is unreliable to evaluate performance by simply using overall accuracy, especially on imbalance data for binary classification.

Table 1: Dataset Description

Description	Values
Duration	01/01/2014 – 31/10/2016 (1034 days)
Data Type	Time Series
Normal Sample	38 757 (91.5%)
Anomaly Sample	3 615 (8.55%)
Total Case	42 372

Table 2: : Confusion Matrix

ACTUAL		PREDICTED	
		POSITIVE	NEGATIVE
	Positive	<i>True Positive (TP)</i>	<i>False Positive (FP)</i>
	Negative	<i>False Negative (FN)</i>	<i>True Negative (TN)</i>

5.1 Evaluation Metrics

[1] claimed that there are several evaluation metrics to be used to evaluate classifiers' performance on imbalance data. It is considered irrelevant to simply use overall accuracy as data skewness exists in this dataset and it is a case of anomaly detection where detecting anomaly and normal users has different costs of misclassification. As it is a form of binary classification problem, the first basis is to come out with the confusion matrix as presented in Table 2

Normal consumer is represented in the negative class, while anomaly consumer is in the positive class. Information can be driven from the confusion matrix as follows:

- TP: Anomaly consumer accurately predicted as Anomaly
- TN: Normal consumer accurately predicted as Normal
- FP: Normal consumer predicted as Anomaly
- FN: Anomaly consumer predicted as Normal

From the confusion matrix, other metrics are generated, namely recall, precision, F-measure and the Area Under Curve (AUC)

- Recall:

$$\frac{TP}{TP + FN}$$

Recall measures the proportion of actual positives out of total actual positives, which expresses the percentage of correctly identified anomaly consumers out of all actual anomaly consumers.

- Precision:

$$\frac{TP}{TP + FP}$$

Precision measures the proportion of actual positives out of total predicted positives, which signifies the proportion of correctly identified anomaly consumers out of all predicted anomaly consumers.

- F1-Score:

$$2 \times \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

This metric seeks for balance between both recall and precision, especially when there is uneven class distribution.

- AUC shows the aggregate measure of performance for all possible classification thresholds. The value range is from 0 to 1, in which a value near to 0 indicates low measure of separability.

5.2 Impact of RUS on selected ML algorithms.

Based on Table 3, RUS has improved the DT performance especially in increasing the proportion of actual anomaly that are correctly identified. DT previously performed very weak in terms of identifying anomaly. Nevertheless, RUS managed to improve the performance. Even so, precision reduced to a very low value, which meant that the percentage of correctness when anomaly was detected was lower because the number of false positive cases increased. For the underfitting issue, the result showed that DT with RUS performed best with the largest training portion; however, it did not confirm that RUS caused underfitting as the 60/40 ratio performed better than 80/20 and 70/30 ratios.

Without resampling, Table 4 confirmed that AdaBoost performed quite better than DT. When applied, RUS helped to increase recall significantly higher in AdaBoost as compared to DT. However, the number of false positives also increased as the precision percentage reduced greatly, which also reduced the F1-score. In contrast, this was worse as compared to the results in DT.

Table 5 shows a huge improvement from Table 3 and Table 4 even without RUS. This signified that neural network could handle imbalance time series data better than DT and its ensemble classifier. When applied with RUS, the performance improved in every aspect except for the precision where false positives also increased. However, the lower training ratio showed that precision still performed better when the training sample was reduced to 60%. It can be concluded that RUS applied in this ANN did not lead to underfitting, but the ANN itself was prone to overfitting.

Based on results shown in Table 6, modifying the ANN into a deep neural network improved the performance, indicating that deep learning had better prediction on time series data. Even without a resampling technique, DANN could predict more anomaly based on the recall percentage. With RUS, DANN performed even better and outperformed all other algorithms used in this study. RUS assisted to increase 16.2% of F1-score. Applying RUS reduced precision in other algorithms since the number of false positives also increased. However, precision in DANN after applying RUS

Table 3: Impact of RUS on DT

Technique	T:T	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)	AUC
DT	90/10	91.08	47.37	2.39	4.55	0.5106
	80/20	91.10	50.77	4.37	8.05	0.5198
	70/30	91.77	53.97	3.24	6.10	0.5149
	60/40	91.59	48.84	1.47	2.86	0.5067
DT with RUS	90/10	82.87	23.04	46.69	30.86	0.6639
	80/20	75.96	16.35	45.69	24.08	0.622
	70/30	80.38	18.90	41.63	25.99	0.6276
	60/40	81.26	20.12	40.65	26.92	0.6284

Table 4: Impact of RUS on AdaBoost

Technique	T:T	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC
AdaBoost	90/10	91.67	63.64	7.67	13.69	0.5363
	80/20	91.54	55.13	5.93	10.71	0.5247
	70/30	92.16	55.49	8.97	15.45	0.5418
	60/40	91.62	55.50	7.70	13.52	0.5356
AdaBoost with RUS	90/10	73.76	18.07	60.11	27.79	0.6756
	80/20	72.59	17.50	60.56	27.16	0.6713
	70/30	71.48	16.82	62.86	26.54	0.6755
	60/40	71.14	16.15	62.95	25.70	0.674

Table 5: Impact of RUS on ANN

Technique	T:T	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC
ANN	90/10	92.54	79.03	13.92	23.67	0.5679
	80/20	92.44	62.66	33.69	43.82	0.6588
	70/30	92.42	57.21	43.62	49.50	0.7029
	60/40	92.74	57.31	50.71	53.81	0.7363
ANN with RUS	90/10	83.95	25.74	49.43	33.85	0.6826
	80/20	86.81	36.38	67.65	47.31	0.7815
	70/30	91.13	48.62	73.01	58.37	0.8292
	60/40	93.53	58.97	73.69	65.51	0.8451

Table 6: Impact of RUS on DANN

Technique	T:T	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)	AUC
DANN	90/10	92.47	65.71	19.60	30.20	0.5934
	80/20	92.44	60.91	38.01	46.80	0.6783
	70/30	93.04	62.64	45.10	52.44	0.713
	60/40	91.29	48.24	61.03	53.89	0.7754
DANN with RUS	90/10	89.71	25.23	46.02	32.60	0.6684
	80/20	93.32	50.33	62.26	55.66	0.7818
	70/30	94.18	60.32	72.37	65.80	0.8397
	60/40	94.97	63.54	78.15	70.09	0.8703

was not highly affected. It showed that this combination was reliable for anomaly detection, whereby increasing the number of actual anomaly detected was not the only concern.

6 DISCUSSION

Anomaly detection using time series data is often challenging with the presence of imbalance data due to the low number of anomaly

cases. RUS is one of the resampling techniques that is fast and reliable to handle imbalance data. This study shows that RUS helped to increase the number of detected anomalies predicted by all classifiers tested. However, it increased the false positive rate at several base classifiers. With the presence of deep learning, the problem can be resolved as deep learning can catch the sequential pattern better, thus creating a clearer boundary between anomaly and normal cases. It was believed that normal cases in time series data constructed similar patterns to each other. For this reason, an undersampling approach was considered as the most appropriate one in this study. Although undersampling is challenged with underfitting issues, this study showed that RUS did not cause underfitting to the classifier for the imbalance time series data that were used, based on the different results on different training/testing ratios.

7 CONCLUSION

Applying RUS on imbalance time series data for anomaly detection increases the rate of true positive and does not clearly cause underfitting issues. This is most probably because normal cases have similar patterns to each other. Neural network portrays better performance than other classifiers although resampling technique was not yet applied. Improving ANN into a deep neural network gave better performance and combining RUS with DANN provided the best results as desired.

In future, it will be useful to observe how the oversampling method tackles the problem with imbalance time series data. Will duplicating data or creating synthetic data overcome the imbalance issue for anomaly detection in a time series data?

ACKNOWLEDGMENTS

The publication of this paper was funded by URND TNB Seeding Fund: U-TE-RD-20-08

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